Fragile Balance of Power

Distributed Fault-Tolerant Seismic Intelligence Platform

Sapienza Università di Roma — Lab of Advanced Programming 2025/2026

The Mission

Year 2038. Amidst peak geopolitical tension, a global network of seismic sensors has been deployed to detect covert military testing. The primary Command Center resides in a strictly neutral region, while data processing centers are scattered globally and vulnerable to attack.

Neutral Region Restriction

The command hub can route data but is strictly prohibited from executing processing logic.

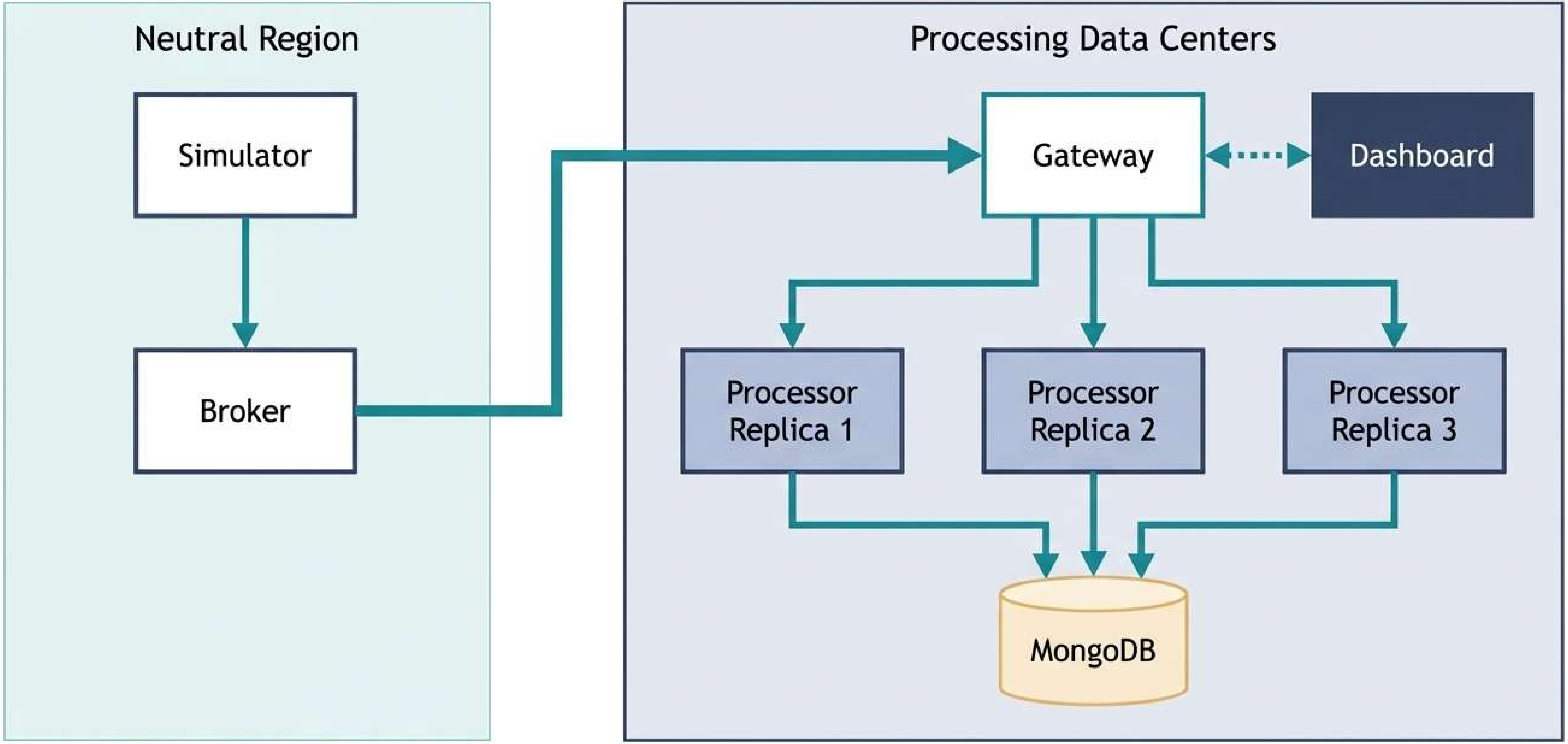
Geographically Distributed DCs

Processing units must operate as independent, distributed nodes.

Partial Failure Tolerance

The system must survive the sudden destruction (shutdown) of individual data centers without dropping telemetry.

System Architecture: The Macro Blueprint



Data Flow: Sensor to Database

1. **Telemetry Generation** — Simulator streams 20 Hz raw sensor data per device.
2. **Socket Initialization** — Broker calls GET /api/devices/ and opens one dedicated WebSocket per sensor.
3. **Stateless Fan-out** — Broker enriches each message with a sensor_id and broadcasts to all available replicas.
4. **Signal Aggregation** — Each replica maintains a discrete 64-sample sliding window per sensor.
5. **FFT Computation** — Fast Fourier Transform is applied; dominant frequency is extracted from the peak magnitude bin.
6. **Secure Persistence** — Event is classified and inserted into MongoDB using a unique compound index on (sensor_id, timestamp).

Fault Tolerance: The Shutdown Sequence

1. Command Injection
Simulator emits an SSE command event containing {'command': 'SHUTDOWN'}.

2. Node Termination
A random replica receives the command and immediately executes process.exit(0).

3. Container Exit Detected
Docker detects the process exit and marks the container as stopped.

4. Gateway Severance
Gateway health checks detect the failure within 5 seconds and sever the node from the round-robin pool.

5. Traffic Absorption
The remaining 2 replicas seamlessly absorb all streaming traffic. The system continues uninterrupted.

Auto-Recovery: Docker 'restart: on-failure' automatically brings the fallen replica back online.

Frequency Classification: Threat Tiers

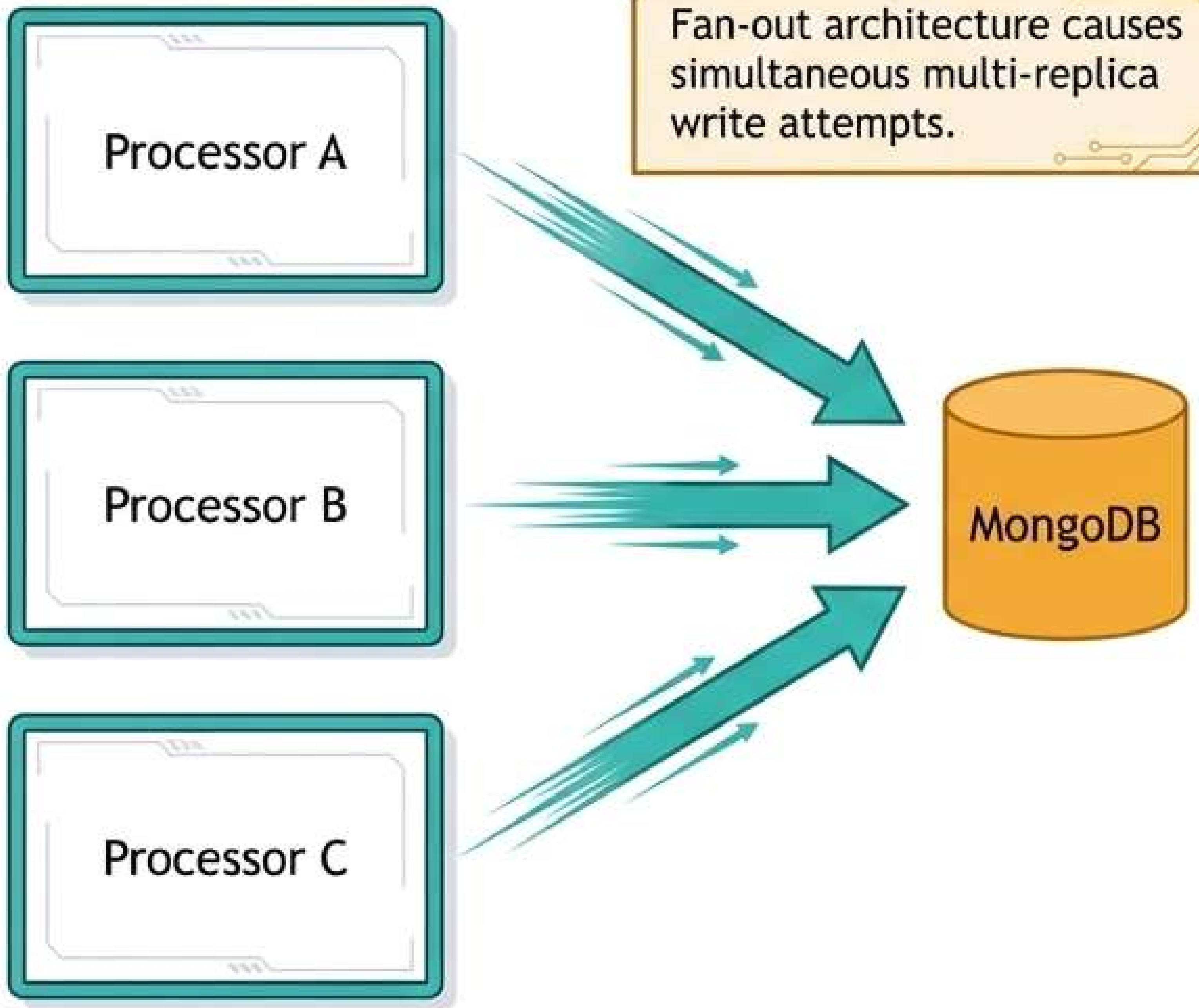
EARTHQUAKE	EXPLOSION	NUCLEAR
0.5 – 3.0 Hz	3.0 – 8.0 Hz	≥ 8.0 Hz

Frequency thresholds are fictional and provided for exam purposes only.

Duplicate-Safe Persistence: Race Condition Resolution

The Conflict

Fan-out architecture causes simultaneous multi-replica write attempts.



The Resolution

```
unique index:
{ sensor_id: 1, timestamp: 1 }

if (err.code !== 11000)
  → log error
else
  → silent discard
```

First writer wins.
All others silently discard.

Dashboard & Monitoring: Command Interface

COMMAND CAPABILITIES

- Live alert feed via Server-Sent Events (SSE).
- Dynamic filtering by event classification type.
- Historical event table querying.
- Live replica health status panel.
- Admin controls (manual node shutdown + synthetic event injection).
- Real-time system log output.

Tech Stack: Foundational Technologies

Node.js 20 ES Modules

Modern, asynchronous runtime foundation.

fft-js

Fast Fourier Transform library for signal processing.

MongoDB 7

Document storage with compound indexing.

ws + eventsource

Bidirectional and unidirectional real-time streams.

Docker + Compose

Containerization and orchestrated networking.

Round-robin Gateway

Application-level load balancing and health checks.

SSE Push

Low-overhead, server-to-client real-time updates.

Sliding Window Analysis

Memory-efficient continuous data buffering.

Key Takeaways

Fan-out without processing

The Broker strictly maintains its neutral status.

Idempotent writes

MongoDB unique indexing effortlessly prevents data duplication.

Health-aware routing

The Gateway dynamically adapts to topology changes in real time.

IaC deployment

A single command orchestrates and starts the entire environment.